

WEEK-WISE PLAN + CHAPTER ARRANGEMENT WITH SUBTOPICS

PHYSICS (Class 12)

1. Electrostatics

Subtopics:

1. Electric Charges & Properties
 2. Conductors, Insulators, Charge Induction
 3. Coulomb's Law
 4. Electric Field & Field Lines
 5. Electric Dipole & Dipole Moment
 6. Electric Flux
 7. Gauss's Law & Applications
 8. Electric Potential & Potential Difference
 9. Equipotential Surfaces
 10. Electric Potential Energy
 11. Capacitance
 12. Combination of Capacitors (Series & Parallel)
 13. Energy Stored in a Capacitor
 14. Dielectrics & Polarization
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2. Current Electricity

Subtopics:

1. Electric Current, Drift Velocity
2. Mobility & Conductivity
3. Ohm's Law
4. Resistivity & Conductance
5. Temperature Dependence of Resistance
6. Electrical Energy & Power
7. Series and Parallel Combination of Resistors
8. Kirchhoff's Laws (KCL & KVL)
9. Wheatstone Bridge
10. Meter Bridge
11. Potentiometer (Principle & Applications)

1. Solid State

Subtopics:

1. Amorphous & Crystalline Solids
 2. Crystal Lattices & Unit Cells
 3. Number of Atoms in Unit Cell
 4. Packing Efficiency
 5. Voids (Tetrahedral & Octahedral)
 6. Density of Unit Cell
 7. Imperfections in Solids
 8. Electrical & Magnetic Properties
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2. Solutions

Subtopics:

1. Types of Solutions
 2. Expressing Concentration (Molarity, Molality, Mass %, ppm)
 3. Solubility
 4. Vapour Pressure of Liquid Solutions
 5. Raoult's Law
 6. Ideal & Non-ideal Solutions
 7. Colligative Properties
 - Relative Lowering of Vapour Pressure
 - Elevation of Boiling Point
 - Depression of Freezing Point
 - Osmotic Pressure
 8. Abnormal Molar Mass & Van't Hoff Factor
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3. Electrochemistry

Subtopics:

1. Electrochemical Cells
2. Galvanic Cell & Electrolytic Cell
3. EMF of Cell
4. Standard Electrode Potential

5. Nernst Equation
 6. Conductance in Electrolytic Solutions
 7. Molar Conductivity
 8. Kohlrausch's Law
 9. Electrolysis & Faraday's Laws
 10. Fuel Cells & Batteries
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MATHEMATICS (Class 12)

1. Relations & Functions

Subtopics:

1. Types of Relations
 2. Functions (Domain, Range, Co-domain)
 3. Types of Functions
 4. Composition of Functions
 5. Invertible Functions
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2. Inverse Trigonometric Functions

Subtopics:

1. Principal Values
 2. Domain & Range
 3. Graphs
 4. Properties
 5. Simple Transformations
 6. Basic Questions
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3. Matrices

Subtopics:

1. Types of Matrices
2. Matrix Addition & Multiplication
3. Scalar Multiplication
4. Zero & Identity Matrix
5. Transpose of Matrix
6. Symmetric & Skew-symmetric Matrices

7. Elementary Operations
 8. Inverse of a Matrix (Elementary Operations method)
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4. Determinants

Subtopics:

1. Determinants of Order 1, 2, 3
 2. Properties of Determinants
 3. Minors & Cofactors
 4. Adjoint
 5. Inverse of Matrix using Determinants
 6. System of Linear Equations (Using Cramer's Rule)
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1-Month (4-Week) Study Plan

Week 1

- Physics: Electrostatics
- Chemistry: Solid State
- Math: Relations & Functions

Week 2

- Physics: Current Electricity
- Chemistry: Solutions
- Math: Inverse Trigonometry

Week 3

- Chemistry: Electrochemistry
- Math: Matrices

Week 4

- Math: Determinants
- Full Revision (all subjects)
- Solve PYQs + NCERT Exemplar + Sample Papers